

Tuesday Warm-up: Ampère's Law, Magnetic Vector Potential

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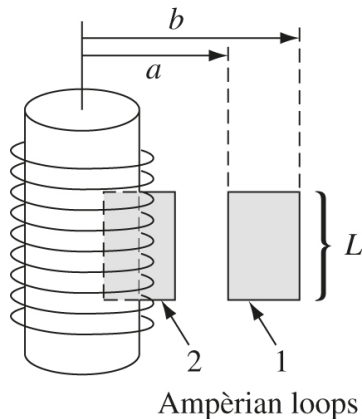


Figure 1: A long solenoid has a constant $\mathbf{B}$ -field inside, but no $\mathbf{B}$ -field outside.

2 Ampère's Law

- (a) Show that if $\nabla \cdot \mathbf{B} = 0$, then we may write $\mathbf{B}$ as the curl of another vector, $\mathbf{A}$. (b) Using Ampère's Law, show that

$$\nabla \times \nabla \times \mathbf{A} = \nabla (\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J} \quad (6)$$

- (c) If we set $\nabla \cdot \mathbf{A} = 0$, then show that $\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}$.

1 Memory Bank

- **Ampère's Law**, integral form:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}} \quad (1)$$

- **Divergence of B-fields:**

$$\nabla \cdot \mathbf{B} = 0 \quad (2)$$

- **Magnetic vector potential:**

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (3)$$

- **Curl of a curl:**

$$\nabla \times \nabla \times \mathbf{A} = \nabla (\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \quad (4)$$

- **Cylindrical curl, z-component:**

$$\nabla \times \mathbf{A} = \frac{1}{s} \left[\frac{\partial}{\partial s} (s A_\phi) - \frac{\partial A_s}{\partial \phi} \right] \hat{\mathbf{z}} \quad (5)$$

- Let the *magnetic flux*, Φ_B be defined like

$$\Phi_B = \int \mathbf{B} \cdot d\mathbf{a} \quad (7)$$

Show that

$$\Phi_B = \oint \mathbf{A} \cdot d\mathbf{l} \quad (8)$$

- Apply the previous result to the long solenoid in Fig. 1. Orient the inner $\mathbf{B}$ -field along the z-axis. (a) What is the magnetic vector potential inside the solenoid? (b) Outside the solenoid? (c) Use the cylindrical curl to obtain the $\mathbf{B}$ -field from the vector potential.¹

¹Note that the vector potential is not zero outside the solenoid.